A REMARK ON OPENAI'S HEIGHT-TWO HAHN–WILSON COUNTEREXAMPLE: EVERY PRIME $p \geq 293$

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Abstract. OpenAI's preprint *Counterexamples to the Hahn–Wilson conjecture at height two* proves its main theorem for every sufficiently large prime. The explicit sufficient condition stated there first holds at $p = 17{,}073{,}437$. We observe that the same argument, with the central window of the digit construction enlarged to the largest radius the proof allows, works for every prime $p \geq 293$. The argument is OpenAI's; we only re-select parameters, and we list every changed line. Developed with AI assistance (Anthropic Claude, Opus 5.5) under [AUTHOR NAME]'s direction; independent adversarial reviews were run by separate AI sessions and are not human peer review.

1. Statement

Line numbers refer to `build/sections/*.tex` of `preprints/Counterexamples-to-the-Hahn-Wilson-conjecture-at-height-two-September-26-2026` in `https://github.com/openai/math` at commit `fd4aeeb`. We write [1] for that preprint. Its main theorem (Theorem 1.1 of [1]) asserts, for every sufficiently large prime p, a connective p-complete spectrum of exact fp-type two outside $\mathrm{Thick}(\mathrm{BP}\langle 2\rangle^{\wedge}_p)$. The only place where the size of p enters beyond $p > 8$ is the parameter condition

$$M \geq 81 \text{ odd}, \qquad p > 16M^2, \qquad 2\lfloor M/8 \rfloor > p^{1/3}, \tag{op:parameters}$$

(`operator.tex` L264–268), together with $R_0 = \lfloor M/8 \rfloor$ (L500). The least prime for which some odd M satisfies (op:parameters) is 17,073,437, attained with $M = 1033$: writing $a = \lfloor M/8 \rfloor$, one needs $8a^3 > 16M^2 \geq 1024a^2$, so $a \geq 129$.

Theorem 1. *Theorem 1.1 of* [1] *holds, with the same proof, for every prime $p \geq 293$.*

For an odd integer $M \geq 11$ put $R_0 = (M-3)/2$, $B_0 = 2R_0 = M-3$, $q = p+1$, and consider

$$\text{(P1)}\quad \frac{p+1}{2M} - \frac{p(M+1)}{p-1} > 1, \qquad \text{(P2)}\quad (M-3)^3 > p,$$

$$\text{(P3)}\quad p > 4M, \qquad \text{(P4)}\quad \Big(\frac{(M-3)p}{M} + 3\Big)(p+1) \leq p^2.$$

Lemma 2. *For every prime $p \geq 293$, the least odd $M \geq 11$ with $(M-3)^3 > p$ satisfies (P1)–(P4).*

Proof. If $M = 11$ then $293 \leq p < 512$; (P1) holds for $M = 11$ exactly when $p \geq 293$ (at $p = 293$ its left side minus one equals $259/803$), and (P3), (P4) are immediate. If $M \geq 13$, minimality gives $p \geq (M-5)^3 \geq 2M(M+3)$. Then $(p+1)/(2M) > M+3$ while $p(M+1)/(p-1) = M+1+(M+1)/(p-1) < M+2$, giving (P1). (P3) is clear, and $p^2 - ((M-3)p/M+3)(p+1) = 3p^2/M - (M-3)p/M - 3p - 3 > 0$ gives (P4). □

Date: October 9, 2026.

2. Proof of Theorem 1: the changed steps

We replace (op:parameters) by "$M \geq 11$ odd and (P1)–(P4)", and $R_0 = \lfloor M/8 \rfloor$ by $R_0 = (M-3)/2$. Then $I = \{s_0 - R_0 + 1, \dots, s_0 + R_0\} = \{2, \dots, M-2\}$ and $I - 1 = \{1, \dots, M-3\}$. We go through every use of these quantities.

Lemma op:coefficient-control (L309–357). The parameters enter only through (op:correction-bound): $qb - r'_m \geq (q/(2M) - \kappa(M+1))X > X$ for $b \geq X/(2M)$ and $m < (M+1)X$, with $\kappa \leq p/(p-1)$ by (op:kappa). This is (P1). Also $M < p$.

Initial depth (L544–564). $d = \lfloor p/M \rfloor$ satisfies $p \leq qd < p^2$, $d \geq p/(2M)$ and $d \leq \delta = (p-1)/2$ whenever $M < p$. At most M digits are protected at one lower label.

First band (L580–603). Column and row lower labels in stage $X = p^r$ differ, below digit $r-1$, by at most the digit diameter of $I \cup (I-1)$ at position $r-2$, plus less than X/p^2 from lower positions. Since $d_{s_0+R_0} - d_{s_0-R_0} \leq 2R_0 p/M + 1$, the bound at L592 becomes $(2R_0 p/M + 3)X/p^2$; the original $(p/4 + 3)$ is the same bound with $2R_0 \leq M/4$. By (P4) this is less than X/q, so same-cluster pairs (and their cyclic complements) fall outside the stage interval $[X/q, X - X/q)$; also it is at most X/p, so a pair of clusters k_1 steps apart has depth in $X(k_1/M \pm 2/p)$. By (P3) the band $k_1 = 1$ lies strictly below the band $k_1 = 2$, and its lower end $X(1/M - 2/p)$ is at least $X/(2M)$, so Lemma op:coefficient-control applies to every depth used.

Retained digits (L605–637). Since $\min I = 2$, column cluster $i \in I$ pairs with row cluster $i - 1 \geq 1$ without cyclic wrap, so retained digits on the two sides agree. Both sides of each adjacent cluster pair have B_0^{r-2} lower labels. The counting argument is unchanged.

Finite crossings (L680–698). The representative of a crossing column label, divided by X, lies within $1/M + 2/p + o(1)$ above $1/2$, and within $1/p$ of $(i + \frac{1}{2})/M$ for its cluster index i. Hence $s_0 \leq i \leq s_0 + 1$ once $p > 3M$, which is within three places of s_0 and inside I because $R_0 \geq 4$. This replaces "since $M \geq 81$" at L694.

Growth (L700–720 and `obstruction.tex` L1082, L1123). One needs $d(I)$ to contain a digit below δ and $d(I-1)$ one above δ ($R_0 \geq 2$), and $\beta = \log_p B_0 > 1/3$, which is (P2).

Rest of the paper. `gluing.tex` and `obstruction.tex` use only the conclusions of Proposition op:operator and $\beta > 1/3$; Lemma obs:growth has constants depending on p but no size condition. The other sections need only $p > 5$, $p > 8$ ($D > 12$), $D > 2$, $D > 3$ and $\gcd(q, p) = 1$. The phrases "sufficiently large prime" in `main.tex` (L57–58, L115, L129) and `consequences.tex` (L14, L172–173, L187) refer only to (op:parameters). □

3. Computational certificate

All checks use exact integer or rational arithmetic.

- `main/hw319_pmin.py` (primary computation): $p_{\min} = 293$ with $(M, R_0) = (11, 4)$; the original condition gives 17,073,437.
- `independent/indep_scan.py` (separate session): re-derivation with exact centers $d_i = \lfloor (2i+1)p/(2M) \rfloor$ and exact band bounds; same value 293.
- `independent/hw319_enumerate.py` (adversarial session): builds the actual surviving label sets of the induction table (op:induction) at $p = 293$, $M = 11$, $R_0 = 4$ for stages $r = 2, 3, 4, 5$ (X up to $2.2 \cdot 10^{12}$; up to 5632^2 label pairs), computes every depth exactly and checks all band, wrap, cut and counting claims. The first-band depth ratio is $b/X \in [0.08687, 0.09465]$, inside $[1/M - 2/p, 1/M + 2/p] = [0.0841, 0.0978]$, with $k_1 = 2$ starting at 0.179. It also verifies (P1)–(P4) for the chosen M at every prime $293 \leq p \leq 2 \cdot 10^6$, and it repeats the stage enumeration at $(509, 11)$, $(521, 13)$, $(1009, 15)$, $(10007, 25)$. Negative controls: $p = 283$ fails (P1); $R_0 = (M-1)/2$ makes I meet an endpoint; $p = 37$ breaks the bands; $M = 9$ fails (P2).

Remark 3. Two sharper variants are recorded in the scripts but not claimed. Reading L689–694 exactly (only $R_0 \geq 2$ is needed) gives 199 (with $M = 9$). Restating Lemma op:coefficient-control with the exact first-band lower end gives 109–167. Both require further changes to the text.

References

[1] OpenAI, *Counterexamples to the Hahn–Wilson conjecture at height two*, OpenAI Math Release preprint, September 26, 2026, `https://github.com/openai/math/tree/main/preprints/Counterexamples-to-the-Hahn-Wilson-conjecture-at-height-two-September-26-2026`.